Chapter 1

Simple Linear Regression: Fit Quality, R^2, and Hypothesis Testing

1.1 The Big Picture

The previous chapter built the simple linear regression model and derived the least squares line

$$\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x. \tag{1.1}$$

The next question is not how to find the line. The next question is:

> Once we have a fitted line, how good is it, and is there statistical evidence that the predictor is useful?

This chapter introduces two related but different ideas.

1. **Fit quality:** How much variation in the response does the fitted line explain? This is measured by R^2.
2. **Statistical evidence:** Is the fitted relationship distinguishable from a zero-slope model after accounting for sample size and noise? This is tested with the F-test or, equivalently in simple linear regression, the slope t-test.

The difference matters. A model can have a high R^2 and still be based on too few observations to trust. A model can have a statistically significant slope but still explain little practical variation. Good analysis reports both fit quality and inferential evidence.

1.2 Review: The Mean-Only Baseline Model

Before using a predictor, we can always estimate a quantitative response Y by its sample mean:

$$\widehat{y}_i = \bar{y}, \qquad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i. \tag{1.2}$$

This is the *mean-only model*. It ignores all predictor values.

The prediction error for observation i under the mean-only model is

$$y_i - \bar{y}. \tag{1.3}$$

The mean squared error of the mean-only model is

$$\frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2. \tag{1.4}$$

This is the denominator-n version of sample variance. Thus, for one-dimensional data, evaluating the mean-only estimate by MSE is essentially evaluating variance.

Math Foundations Connection. A response column $(y_1, \ldots, y_n)^T$ is a vector in $\mathbb{R}^n$, and the mean-only fitted vector can be written as $\bar{y}\mathbf{1}$. The sums of squared deviations below are squared Euclidean norms; see the Math Foundations textbook, Chapters 2, 3, and 5.

1.3 Adding a Predictor

When we observe paired data

$$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n), \tag{1.5}$$

we can compare two strategies:

1. predict all responses using $\bar{y}$;
2. predict responses using the fitted regression line

$$\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i. \tag{1.6}$$

If X is useful, then the fitted line should have smaller squared prediction error than the mean-only model. If X is useless, then the fitted line will look like a horizontal line with slope near zero, and it will not improve much over the mean-only model.

This is the core intuition behind R^2 and the F-test:

> Compare the error from the mean-only model to the error from the fitted regression model.

1.4 Three Sums of Squares

Let

$$e_i = y_i - \widehat{y}_i \tag{1.7}$$

be the residual from the fitted regression line.

1.4.1 Total Sum of Squares

The *total sum of squares* measures how much the response varies around its sample mean:

$$\mathrm{TSS} = \sum_{i=1}^{n} (y_i - \bar{y})^2. \tag{1.8}$$

This is the squared error from the mean-only model.

1.4.2 Residual Sum of Squares

The *residual sum of squares* measures how much variation remains after fitting the regression line:

$$\mathrm{RSS} = \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2 = \sum_{i=1}^{n} e_i^2. \tag{1.9}$$

This is the squared error from the fitted regression model.

1.4.3 Regression Sum of Squares

The *regression sum of squares*, sometimes called explained sum of squares, measures how much variation the model explains relative to the mean-only baseline:

$$\mathrm{RegSS} = \mathrm{TSS} - \mathrm{RSS}\,. \tag{1.10}$$

When the regression model includes an intercept, least squares gives the decomposition

$$\boxed{\mathrm{TSS} = \mathrm{RegSS} + \mathrm{RSS}\,.} \tag{1.11}$$

Math Foundations Connection. The decomposition in (1.11) is a geometry statement. It relies on the residual vector being orthogonal to the fitted-value direction after centering. Dot products, norms, and orthogonality are covered in the Math Foundations textbook, Chapters 2, 3, and 5.

1.5 R^2: Proportion of Variation Explained

The coefficient of determination is

$$\boxed{R^2 = \frac{\mathrm{TSS} - \mathrm{RSS}}{\mathrm{TSS}} = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}.} \tag{1.12}$$

The numerator $\mathrm{TSS} - \mathrm{RSS}$ is the reduction in squared error achieved by using the predictor. The denominator TSS is the squared error we had before using the predictor.

Thus,

> R^2 is the proportion of the response variation explained by the regression model.

If $R^2 = 0$, then the fitted model does no better than the mean-only model. If $R^2 = 1$, then the residuals are all zero and the fitted line passes exactly through every observed point.

For simple linear regression with an intercept,

$$0 \le R^2 \le 1. \tag{1.13}$$

This interval property depends on comparing an intercept-containing regression model to the mean-only baseline. Some nonstandard models, such as no-intercept regressions, can produce different behavior.

1.6 Interpreting R^2

Suppose two models are used to predict pediatric weight.

- Model A uses wrist circumference and has $R^2 = 0.25$.
- Model B uses height and has $R^2 = 0.50$.

Then Model B explains twice as much of the observed weight variation as Model A, at least in the data used to fit the models.

The phrase *explains variation* should be read carefully. It does not automatically imply causation. It means that the fitted line reduces squared prediction error relative to the mean-only model.

1.7 R^2 and Correlation in Simple Linear Regression

For simple linear regression with an intercept, R^2 is equal to the squared sample correlation between x and y:

$$\boxed{R^2 = r_{xy}^2.} \tag{1.14}$$

To see why, define centered values

$$x_i^c = x_i - \bar{x}, \qquad y_i^c = y_i - \bar{y}. \tag{1.15}$$

Let

$$S_{xx} = \sum_{i=1}^{n} (x_i^c)^2, \qquad S_{yy} = \sum_{i=1}^{n} (y_i^c)^2, \qquad S_{xy} = \sum_{i=1}^{n} x_i^c y_i^c. \tag{1.16}$$

The least squares slope is

$$\widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}}. \tag{1.17}$$

The centered fitted values satisfy

$$\widehat{y}_i - \bar{y} = \widehat{\beta}_1 (x_i - \bar{x}). \tag{1.18}$$

Therefore, the regression sum of squares is

$$\begin{aligned}
\text{RegSS} &= \sum_{i=1}^{n} (\widehat{y}_i - \bar{y})^2 && (1.19)\\
&= \sum_{i=1}^{n} \widehat{\beta}_1^2 (x_i - \bar{x})^2 && (1.20)\\
&= \widehat{\beta}_1^2 S_{xx} && (1.21)\\
&= \left(\frac{S_{xy}}{S_{xx}}\right)^2 S_{xx} && (1.22)\\
&= \frac{S_{xy}^2}{S_{xx}}. && (1.23)
\end{aligned}$$

Thus,

$$\begin{aligned}
R^2 &= \frac{\text{RegSS}}{\text{TSS}} && (1.24)\\
&= \frac{S_{xy}^2/S_{xx}}{S_{yy}} && (1.25)\\
&= \frac{S_{xy}^2}{S_{xx}S_{yy}} && (1.26)\\
&= \left(\frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}\right)^2 && (1.27)\\
&= r_{xy}^2. && (1.28)
\end{aligned}$$

This equality is useful in SLR. It does not directly generalize to multiple linear regression as a single ordinary correlation between one predictor and the response.

1.8 Limitations of R^2

R^2 is useful, but it does not tell us everything.

1.8.1 A Perfect Fit Can Be Misleading

With two data points and a line with two parameters, we can fit the data perfectly as long as the two x values are different. Then

$$\text{RSS} = 0 \qquad \Rightarrow \qquad R^2 = 1. \tag{1.29}$$

This does not mean the model will predict future data well. It means only that a line can pass through two points.

1.8.2 R^2 Does Not Measure Statistical Evidence

R^2 measures in-sample fit. It does not directly answer:

- Is the slope distinguishable from zero?
- How much sampling uncertainty is in $\widehat{\beta}_1$?
- How noisy are the observations?
- How many observations were used?

These questions require hypothesis tests, confidence intervals, and residual diagnostics.

1.8.3 R^2 Does Not Check Model Assumptions

A high R^2 does not prove that the linear model assumptions are valid. Residual plots may still reveal curvature, nonconstant variance, outliers, or dependence among observations.

1.9 From Fit Quality to Hypothesis Testing

Fit quality asks:

> How much did the line reduce squared error compared to the mean-only model?

Hypothesis testing asks:

> Is that reduction large enough, relative to residual noise and sample size, to count as statistical evidence that the slope is not zero?

For SLR, the inferential question is usually

$$H_0 : \beta_1 = 0 \qquad \text{versus} \qquad H_A : \beta_1 \neq 0. \tag{1.30}$$

The null hypothesis says that X does not improve the conditional mean of Y through a linear slope. The alternative says that the slope is nonzero.

Math Foundations Connection. Hypotheses are mathematical statements, and hypothesis testing depends on understanding logical alternatives and negation. Basic logical statements, negation, and truth-table reasoning are covered in the Math Foundations textbook, Chapter 1.

1.10 The F-Statistic

The F-statistic compares the mean-only model to the fitted regression model.

For a general nested-model comparison,

$$F = \frac{(\mathrm{SS}_{\mathrm{null}} - \mathrm{SS}_{\mathrm{full}})\,/(p_{\mathrm{full}} - p_{\mathrm{null}})}{\mathrm{SS}_{\mathrm{full}}\,/(n - p_{\mathrm{full}})}, \tag{1.31}$$

where:

- $\mathrm{SS}_{\mathrm{null}}$ is the squared error from the simpler model;

- $\mathrm{SS}_{\mathrm{full}}$ is the squared error from the larger model;
- p_{null} is the number of parameters in the simpler model;
- p_{full} is the number of parameters in the larger model.

For simple linear regression, the null model is the mean-only model, so

$$p_{\mathrm{null}} = 1, \qquad \mathrm{SS}_{\mathrm{null}} = \mathrm{TSS}\,. \tag{1.32}$$

The full model is the fitted line, so

$$p_{\mathrm{full}} = 2, \qquad \mathrm{SS}_{\mathrm{full}} = \mathrm{RSS}\,. \tag{1.33}$$

Therefore,

$$F = \frac{(\mathrm{TSS} - \mathrm{RSS})/(2-1)}{\mathrm{RSS}\,/(n-2)} \tag{1.34}$$

$$= \boxed{\frac{\mathrm{TSS} - \mathrm{RSS}}{\mathrm{RSS}\,/(n-2)}}\,. \tag{1.35}$$

The numerator is the improvement over the mean-only model. The denominator is the residual variation per leftover degree of freedom.

Math Foundations Connection. The phrase *leftover degrees of freedom* is a dimension idea. The residual variation lives in directions left over after fitting the intercept and slope. Linear independence, bases, and the fact that a linearly independent set of n-vectors can have at most n elements are introduced in the Math Foundations textbook, Chapters 2, 5, and 6.

1.11 The F-Statistic in Terms of R^2

Because

$$R^2 = \frac{\mathrm{TSS} - \mathrm{RSS}}{\mathrm{TSS}}, \tag{1.36}$$

we can rewrite the SLR F-statistic as

$$F = \frac{\mathrm{TSS} - \mathrm{RSS}}{\mathrm{RSS}\,/(n-2)} \tag{1.37}$$

$$= \frac{R^2\,\mathrm{TSS}}{(1-R^2)\,\mathrm{TSS}\,/(n-2)} \tag{1.38}$$

$$= \boxed{\frac{R^2}{1-R^2}(n-2).} \tag{1.39}$$

This formula explains why sample size matters. If two simple linear regressions have the same R^2, the one with more observations will have a larger F-statistic. More data can make the same proportion of explained variation more statistically detectable.

1.12 The F-Distribution and the p-Value

Under Assumption 3 from Chapter 5—the Gaussian error model—and the null hypothesis $H_0 : \beta_1 = 0$,

$$F \sim F_{1,n-2}. \tag{1.40}$$

The first degree of freedom is the numerator degree of freedom:

$$p_{\text{full}} - p_{\text{null}} = 2 - 1 = 1. \tag{1.41}$$

The second degree of freedom is the residual degree of freedom:

$$n - p_{\text{full}} = n - 2. \tag{1.42}$$

After calculating the observed statistic F_{obs}, the p-value is

$$p\text{-value} = \Pr\left(F_{1,n-2} \geq F_{\text{obs}} \mid H_0\right). \tag{1.43}$$

A small p-value means that, if the true slope were zero and the model assumptions held, an improvement this large would be unlikely by random sampling variation alone.

1.13 Why the Assumptions Matter

The F-test relies on the same three-step regression assumption ladder introduced in Chapter 5.

1. Assumption 1 (mean-zero errors) supports the interpretation of the line as the conditional mean.
2. Assumption 2 (constant variance and uncorrelated errors) supports the standard error and variance calculations.
3. Assumption 3 (Gaussian errors) supports exact finite-sample t and F distributions.

If these assumptions are badly violated, the p-value may not have the advertised interpretation. This is why residual analysis is not optional.

1.14 The Slope t-Test

Most software output also reports a t-test for the slope. The slope test is

$$H_0 : \beta_1 = 0 \qquad \text{versus} \qquad H_A : \beta_1 \neq 0. \tag{1.44}$$

The residual variance estimator is

$$\widehat{\sigma}^2 = \frac{\text{RSS}}{n-2}. \tag{1.45}$$

The standard error of the slope is

$$\text{SE}(\widehat{\beta}_1) = \sqrt{\frac{\widehat{\sigma}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}} = \sqrt{\frac{\widehat{\sigma}^2}{S_{xx}}}. \tag{1.46}$$

The t-statistic is

$$t_{\text{obs}} = \frac{\widehat{\beta}_1}{\text{SE}(\widehat{\beta}_1)}. \tag{1.47}$$

Under the null hypothesis and Assumption 3 (Gaussian errors),

$$t_{\text{obs}} \sim t_{n-2}. \tag{1.48}$$

1.15 Why the SLR F-Test and Slope t-Test Agree

In simple linear regression with an intercept,

$$\boxed{F = t^2.} \tag{1.49}$$

To see why, recall that

$$\text{RegSS} = \widehat{\beta}_1^2 S_{xx}. \tag{1.50}$$

Also,

$$\text{SE}(\widehat{\beta}_1)^2 = \frac{\widehat{\sigma}^2}{S_{xx}}, \qquad \widehat{\sigma}^2 = \frac{\text{RSS}}{n-2}. \tag{1.51}$$

Thus,

$$\begin{aligned} t^2 &= \frac{\widehat{\beta}_1^2}{\text{SE}(\widehat{\beta}_1)^2} && (1.52)\\ &= \frac{\widehat{\beta}_1^2}{\widehat{\sigma}^2 / S_{xx}} && (1.53)\\ &= \frac{\widehat{\beta}_1^2 S_{xx}}{\widehat{\sigma}^2} && (1.54)\\ &= \frac{\text{RegSS}}{\text{RSS}\,/(n-2)} && (1.55)\\ &= F. && (1.56) \end{aligned}$$

Therefore, in SLR, the overall F-test for the model and the two-sided t-test for the slope produce the same conclusion. In multiple linear regression, the overall F-test and individual coefficient t-tests answer different questions.

1.16 Worked Example

Consider five observations:

x_i	y_i
1	2
2	3
3	5
4	4
5	6

The least squares line is

$$\widehat{y} = 1.3 + 0.9x. \tag{1.57}$$

The fitted values and residuals are:

i	x_i	y_i	$\widehat{y}_i$	e_i
1	1	2	2.2	−0.2
2	2	3	3.1	−0.1
3	3	5	4.0	1.0
4	4	4	4.9	−0.9
5	5	6	5.8	0.2

The sample mean is

$$\bar{y} = 4. \tag{1.58}$$

The total sum of squares is

$$\text{TSS} = (2-4)^2 + (3-4)^2 + (5-4)^2 + (4-4)^2 + (6-4)^2 = 10. \tag{1.59}$$

The residual sum of squares is

$$\begin{aligned} \text{RSS} &= (-0.2)^2 + (-0.1)^2 + (1.0)^2 + (-0.9)^2 + (0.2)^2 && (1.60) \\ &= 1.9. && (1.61) \end{aligned}$$

Therefore,

$$R^2 = 1 - \frac{1.9}{10} = 0.81. \tag{1.62}$$

The model explains 81% of the observed variation in y.

The F-statistic is

$$\begin{aligned} F &= \frac{\text{TSS} - \text{RSS}}{\text{RSS}/(n-2)} && (1.63) \\ &= \frac{10 - 1.9}{1.9/3} && (1.64) \\ &\approx 12.79. && (1.65) \end{aligned}$$

Under $H_0 : \beta_1 = 0$, this is compared to an $F_{1,3}$ distribution. Equivalently, the slope t-statistic is

$$t = \sqrt{F} \approx 3.58, \tag{1.66}$$

with the sign determined by the sign of $\widehat{\beta}_1$. Since the slope is positive, $t \approx 3.58$.

1.17 How to Read Software Output

Regression software usually reports both fit-quality and hypothesis-testing information.

1.17.1 Fit Quality Fields

Common fields include:

- **R-squared:** the R^2 value;
- **Residual standard error** or **scale**: an estimate of residual noise;
- **F-statistic:** the model-versus-null comparison;
- **Prob(F-statistic):** the p-value for the F-test.

1.17.2 Coefficient Table Fields

For the predictor row, common fields include:

- **coef:** $\widehat{\beta}_1$;
- **std err:** $\mathrm{SE}(\widehat{\beta}_1)$;
- **t:** the slope t-statistic;
- **P¿—t—:** the two-sided p-value for $H_0 : \beta_1 = 0$;
- **confidence interval:** an interval estimate for the slope.

In SLR, the p-value in the predictor row and the p-value for the model F-statistic correspond to the same slope question.

1.18 Decision Template for SLR Fit and Significance

When reporting a simple linear regression, use the following template.

1. **State the fitted model:**
$$\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x. \tag{1.67}$$
2. **Interpret the slope with units.**
3. **Report R^2:** say what proportion of response variation is explained by the predictor.
4. **State the hypothesis test:**
$$H_0 : \beta_1 = 0 \qquad \text{versus} \qquad H_A : \beta_1 \neq 0. \tag{1.68}$$
5. **Report the test statistic and p-value.**
6. **Translate the conclusion into context.**
7. **Check residuals before overclaiming.**
8. **Avoid causal language unless the design supports causation.**

Coding companion reference. Package-specific Python/R commands for computing R^2, F-statistics, p-values, and related output have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the conceptual guidance for reading software output.

1.19 Common Mistakes

1. **Treating high R^2 as proof of causation.** R^2 is about explained variation, not causal mechanism.
2. **Treating high R^2 as proof of validity.** A model can have high R^2 and still violate assumptions.
3. **Ignoring sample size.** The same R^2 can produce different statistical evidence depending on n.
4. **Confusing residual variance with response variance.** TSS measures response variation before the model; RSS measures variation left after the model.
5. **Forgetting degrees of freedom.** In SLR, residual variance uses denominator $n - 2$ because two parameters are estimated.
6. **Thinking $R^2 = 1$ always means a useful model.** Two points can give a perfect line, but not a reliable model.
7. **Reading p-values without context.** A small p-value does not tell you effect size, practical importance, or model adequacy.

1.20 Chapter Summary

The mean-only model predicts every response using $\bar{y}$. Its squared error is the total sum of squares,

$$\mathrm{TSS} = \sum_{i=1}^{n} (y_i - \bar{y})^2. \tag{1.69}$$

The fitted regression model predicts each response with

$$\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i. \tag{1.70}$$

Its squared error is the residual sum of squares,

$$\mathrm{RSS} = \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2. \tag{1.71}$$

The coefficient of determination is

$$R^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}. \tag{1.72}$$

It measures the proportion of observed response variation explained by the fitted line.

The F-test compares the fitted line to the mean-only model:

$$F = \frac{\mathrm{TSS} - \mathrm{RSS}}{\mathrm{RSS}\,/(n-2)}. \tag{1.73}$$

Under $H_0 : \beta_1 = 0$ and Assumption 3 (Gaussian errors),

$$F \sim F_{1,n-2}. \tag{1.74}$$

In simple linear regression with an intercept, the slope t-test and model F-test are equivalent:

$$F = t^2. \tag{1.75}$$

The central distinction is this:

> R^2 measures how much the model fits the observed data; the F-test and t-test measure whether the observed improvement is statistically distinguishable from a zero-slope model under the same regression assumption ladder.

1.21 Math Foundations Source Map

This source map records the Math Foundations support used in this chapter and where those ideas appear in the completed Math Foundations textbook.

Chapter 6 use	**Math Foundations connection**	**MF textbook location**
Mean-only fitted vector	$\mathbb{R}^n$ vector notation and the ones vector $\mathbf{1}$	Ch02 Secs. 2.1, 2.4
TSS and RSS as squared lengths	Dot products, $\mathbf{a}^T\mathbf{a} = \sum_i a_i^2$, and Euclidean norm $\|\mathbf{x}\| = \sqrt{\mathbf{x}^T\mathbf{x}}$	Ch05 Secs. 5.1, 5.3, 5.5
Sum-of-squares decomposition	Dot products, norms, and orthogonality of fitted and residual components	Ch02 Sec. 2.6; Ch05 Secs. 5.1, 5.3, 5.5, 5.8; Ch12 Sec. 12.12
Residual degrees of freedom	Linear independence, basis size, and leftover dimension intuition	Ch02 Sec. 2.6; Ch06 Secs. 6.3, 6.4, 6.5
Hypothesis statements	Logic statements, negation, and truth-table reasoning as the foundation for null-versus-alternative statements	Ch01 Sec. 1.1
Software/design-matrix bridge	Matrix notation, transpose notation, and matrix-vector multiplication	Ch07 Secs. 7.1, 7.4, 7.6.1, 7.6.2

1.22 Practice and Workflow

Focused Study Checklist

Use this as a final check after analyzing fit quality and statistical evidence.

1. Can you identify the mean-only baseline and compute TSS?
2. Can you compute residuals, RSS, and $R^2 = 1 - \text{RSS} / \text{TSS}$?
3. Can you explain R^2 as fit quality, not as proof that the model is useful or causal?
4. Can you state the slope test hypotheses $H_0 : \beta_1 = 0$ and $H_A : \beta_1 \neq 0$?
5. Can you connect the model F-test to the slope t-test in SLR through $F = t^2$?
6. Can you conclude in context while separating fit, statistical evidence, and residual diagnostics?